AMEEMAW

AI Mamma-Echography Educator & Empathetic Mentor

for Awareness & Wellness

Technical Report

Breast Ultrasound Classification with Explainable AI

ASIAN INSTITUTE OF MANAGEMENT
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Table of Contents

Abstract 3
1. Introduction 4
1.1 Problem Statement 4
1.2 Objectives 4
1.3 Success Metrics and KPI Rationale 5
2. Methods 6
2.1 Dataset 6
2.2 Data Preprocessing 6
2.3 Feature Engineering 7
2.4 Model Architecture 7
2.5 Explainability Methods 8
2.6 Bias Audit Framework 8
3. Results 10
3.1 Model Performance Comparison 10
3.2 Deployed Model Performance (ResNet-50 3-Class) 10
3.3 Explainability Analysis 11
3.3.1 Grad-CAM Visual Explanations 11
3.3.2 SHAP Feature Importance Analysis 12
3.4 Bias Audit Results 14
3.4.1 Critical Finding: Small Malignant Lesion Detection 14
3.4.2 Size and Brightness Bias Summary 14
3.4.3 Evaluation Against Success Metrics 14
4. Discussion 16
4.1 Key Findings 16
4.2 Limitations 16
4.3 Ethical Considerations and Safeguards 17
4.4 Future Work 17
5. Conclusion 18
References 19
Appendix A: Complete Model Comparison 20

Abstract

AMEEMAW (AI Mamma-Echography Educator & Empathetic Mentor for Awareness & Wellness) is an educational artificial intelligence system designed to classify breast ultrasound images into three categories: normal, benign, and malignant. Breast cancer remains one of the leading causes of cancer-related mortality among women worldwide, yet early detection through screening significantly improves survival rates—from 27% for late-stage diagnosis to 99% for early-stage detection (American Cancer Society, 2024). This project addresses the critical need for accessible breast health education tools while demonstrating responsible AI development practices.

Using the publicly available BUSI dataset comprising 780 ultrasound images (Al-Dhabyani et al., 2020), this study trained and compared 15 machine learning models across two paradigms: deep learning convolutional neural networks (CNNs) and traditional tabular models using radiomics features. The best-performing model, Logistic Regression trained on 102 radiomics features, achieved 93.54% F1-score on binary classification. For the deployed 3-class CNN system, ResNet-50 achieved 84.62% accuracy with 83.9% malignant recall.

Comprehensive explainability analysis using Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive exPlanations (SHAP) revealed that the model focuses on clinically relevant regions consistent with BI-RADS criteria. Systematic bias auditing identified critical limitations, including only 50% recall on small malignant lesions—a finding with significant clinical implications for early-stage cancer detection. The system incorporates Nana, a generative AI companion powered by Claude Haiku API, providing empathetic, plain-language explanations to support patient education. This report details the methodology, findings, limitations, and recommendations for responsible educational deployment.

Keywords: *breast ultrasound, deep learning, explainable AI, Grad-CAM, SHAP, bias auditing, medical imaging, responsible AI*

1. Introduction

1.1 Problem Statement

Breast cancer is the most commonly diagnosed cancer among women globally, with approximately 2.3 million new cases annually (World Health Organization, 2023). The disease accounts for 685,000 deaths per year worldwide, making it the leading cause of cancer-related mortality in women. However, the prognosis varies dramatically based on the stage at diagnosis: the 5-year survival rate exceeds 99% for localized early-stage breast cancer but drops to only 27% for distant metastatic disease (American Cancer Society, 2024). This stark contrast underscores the critical importance of early detection.

Ultrasound imaging serves as a vital complementary screening tool, particularly for women with dense breast tissue where mammography sensitivity decreases (Kolb et al., 2002). Unlike mammography, ultrasound provides real-time imaging without ionizing radiation, making it especially valuable for younger women and those requiring frequent follow-up. However, accurate interpretation of breast ultrasound images requires specialized training and expertise. The American College of Radiology's Breast Imaging Reporting and Data System (BI-RADS) provides standardized assessment categories, but applying these criteria consistently requires years of experience.

This expertise gap creates significant accessibility barriers. Many communities, particularly in underserved areas, lack access to specialized breast imaging radiologists. Furthermore, patients who receive ultrasound results often lack the context to understand what their BI-RADS scores mean, leading to anxiety and confusion. AMEEMAW addresses these challenges by providing an educational AI tool that can help train future specialists, assist current practitioners in pattern recognition, and support patients in understanding their results through compassionate, plain-language explanations.

It is essential to emphasize that AMEEMAW is designed as <u>an educational tool, not a diagnostic system</u>. The tool does not replace the need for qualified radiologists and healthcare professionals. Rather, it aims to support the training pipeline and improve health literacy around breast ultrasound interpretation.

1.2 Objectives

The primary objectives of this project are:

1. Develop and compare multiple machine learning models for breast ultrasound classification into three categories: normal, benign, and malignant
2. Implement explainability techniques (Grad-CAM for CNNs, SHAP for tabular models) to ensure model transparency and validate clinical relevance
3. Conduct comprehensive bias auditing across clinically relevant subgroups to identify performance disparities
4. Create an accessible, user-friendly educational interface with a generative AI companion for patient support
5. Document all limitations and ethical considerations for responsible deployment

1.3 Success Metrics and KPI Rationale

Defining appropriate success metrics for medical AI applications requires careful consideration of clinical context, ethical implications, and stakeholder needs. The following key performance indicators (KPIs) were selected based on established best practices in medical imaging AI and the specific requirements of breast cancer screening.

Technical KPIs

1. **F1-Score > 80%:** The F1-score was chosen as the primary metric because it balances precision and recall, which is critical in imbalanced medical datasets. The BUSI dataset exhibits significant class imbalance (56% benign, 27% malignant, 17% normal), making accuracy alone misleading. An 80% threshold was set based on comparable studies in breast ultrasound classification (Han et al., 2017).
2. **Malignant Recall > 90%:** In cancer screening, the cost of a false negative (missing a cancer) far exceeds the cost of a false positive (unnecessary follow-up). A missed cancer diagnosis can lead to delayed treatment and significantly worse outcomes. The 90% target aligns with the US FDA guidance for computer-aided detection (CAD) systems, which emphasizes sensitivity for malignancy detection (FDA, 2020). This metric was derived from clinical standards where screening mammography aims for sensitivity above 85% (Lehman et al., 2017).
3. **AUC-ROC > 0.90:** The Area Under the Receiver Operating Characteristic curve provides a threshold-independent measure of model discrimination ability. An AUC of 0.90 indicates excellent discrimination and is consistent with published benchmarks for breast ultrasound CAD systems (Cheng et al., 2010).

Educational and Ethical KPIs

- **Transparency:** All model predictions must be accompanied by explainability visualizations
- **Bias Documentation:** Systematic analysis of model performance across subgroups with full disclosure of disparities
- **Clear Disclaimers:** Prominent messaging that the tool is for education only, not clinical diagnosis
- **Appropriate Referral:** Consistent encouragement to consult healthcare professionals

2. Methods

2.1 Dataset

This study utilized the Breast Ultrasound Images (BUSI) dataset, a publicly available collection released by Al-Dhabyani et al. (2020) and hosted on Kaggle under a Creative Commons Attribution 4.0 (CC BY 4.0) license. The dataset was collected at Baheya Hospital for Early Detection and Treatment of Women's Cancer in Cairo, Egypt, from women aged 25 to 75 years.

Dataset Characteristics:

Total Images	780 grayscale PNG images
Classes	Normal (133, 17.1%), Benign (437, 56.0%), Malignant (210, 26.9%)
Image Dimensions	Variable (approximately 500×500 pixels average)
Segmentation Masks	Available for benign and malignant cases
Patient Demographics	Women aged 25-75 years

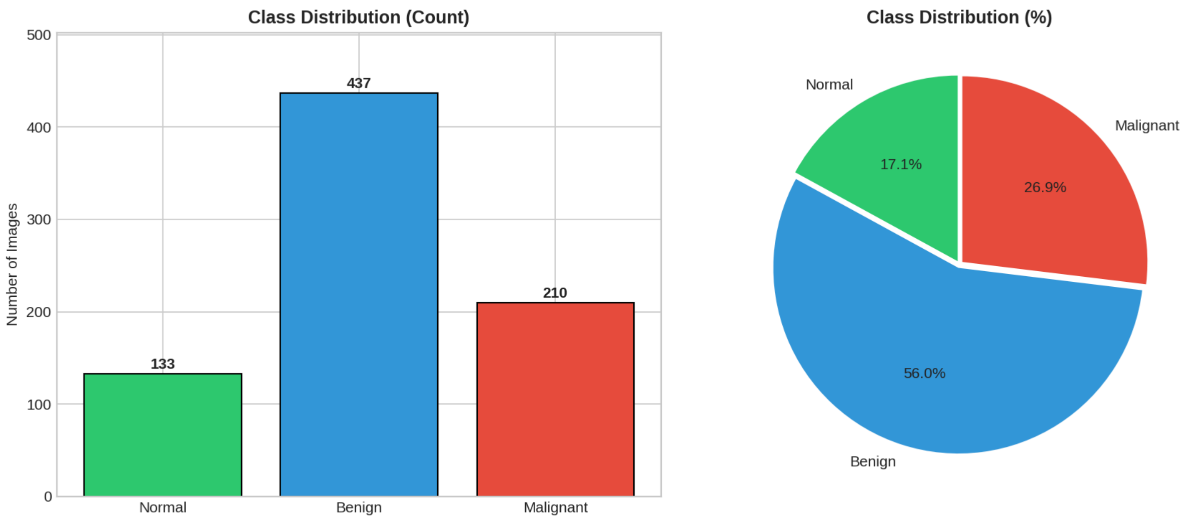

Figure 1: Class Distribution in BUSI Dataset

The dataset exhibits significant class imbalance with benign cases comprising over half of the samples.

2.2 Data Preprocessing

A standardized preprocessing pipeline was implemented to prepare images for both CNN and radiomics analysis. The dataset was split into training (70%, n=546), validation (15%, n=117), and test (15%, n=117) sets using stratified random sampling to preserve class proportions across splits.

Image Preprocessing Pipeline:

1. **Resizing:** All images resized to 224×224 pixels using bilinear interpolation, the standard input size for ImageNet-pretrained models

2. **Channel Conversion:** Grayscale images converted to 3-channel RGB for compatibility with pretrained networks
3. **Normalization:** ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225) applied for transfer learning
4. **Data Augmentation (training only):** Random horizontal flip (p=0.5), rotation (±15°), brightness adjustment (±20%), and contrast adjustment (±20%)

To address class imbalance, weighted random sampling was employed during training, assigning higher sampling probability to minority classes (normal, malignant) proportional to their inverse frequency.

2.3 Feature Engineering

For traditional machine learning models, 102 radiomics features were extracted from each image using the PyRadiomics library (van Griethuysen et al., 2017). Radiomics refers to the extraction of quantitative features from medical images that may not be perceptible to the human eye but contain valuable diagnostic information.

Feature Categories:

Category	Count	Description
Shape (2D)	10	Geometric properties: sphericity, elongation, perimeter
First-Order Statistics	18	Intensity histogram: mean, variance, entropy, skewness
GLCM	24	Gray Level Co-occurrence Matrix texture features
GLRLM	16	Gray Level Run Length Matrix features
GLSZM	16	Gray Level Size Zone Matrix features
NGTDM	5	Neighborhood Gray Tone Difference Matrix
GLDM	13	Gray Level Dependence Matrix features

All radiomics features were standardized using z-score normalization (StandardScaler), with parameters fit on the training set and applied to validation and test sets to prevent data leakage.

2.4 Model Architecture

Convolutional Neural Networks (Deep Learning)

Five CNN architectures were evaluated, all initialized with ImageNet pretrained weights and fine-tuned for breast ultrasound classification:

- **ResNet-50** (He et al., 2016): 50-layer residual network with skip connections
- **EfficientNet-B0** (Tan & Le, 2019): Compound-scaled network optimizing depth, width, and resolution
- **VGG-16** (Simonyan & Zisserman, 2015): Classical 16-layer deep network
- **MobileNetV2** (Sandler et al., 2018): Lightweight architecture with inverted residuals
- **Custom CNN:** Baseline 5-layer convolutional network trained from scratch

Each model was trained with the Adam optimizer (learning rate = 1×10^{-4}), cross-entropy loss with class weights inversely proportional to class frequencies, and early stopping with patience of 10 epochs monitoring validation loss. Models were evaluated on both 3-class (normal/benign/malignant) and 2-class (benign/malignant) classification tasks.

Traditional Machine Learning (Tabular Models)
Five traditional machine learning algorithms were trained on the 102 radiomics features for 2-class (benign/malignant) classification:

- **Logistic Regression**
- **XGBoost**
- **Random Forest**
- **Support Vector Machine (RBF kernel)**
- **Multi-Layer Perceptron**

Hyperparameter tuning was performed using 5-fold cross-validation with grid search.

2.5 Explainability Methods

Model interpretability is essential for medical AI applications to build clinician trust and validate that models learn clinically meaningful patterns rather than spurious correlations (Rudin, 2019). Two complementary explainability techniques were employed:

Gradient-weighted Class Activation Mapping (Grad-CAM)
Grad-CAM (Selvaraju et al., 2017) generates visual explanations by computing the gradient of the predicted class score with respect to feature maps in the final convolutional layer. The technique produces heatmaps highlighting image regions most influential in the model's decision. For ResNet-50, Grad-CAM was applied to the output of layer4 (the final residual block). Heatmaps were overlaid on original images using a jet colormap, with warmer colors (red/yellow) indicating regions of higher importance.

A 'focus score' metric was computed to quantify how concentrated the model's attention was on lesion regions versus background, using the ratio of activation intensity within the lesion mask to total activation.

SHapley Additive exPlanations (SHAP)
SHAP (Lundberg & Lee, 2017) provides feature-level explanations based on cooperative game theory. Each feature's contribution to a prediction is computed as a Shapley value, representing the average marginal contribution across all possible feature coalitions. SHAP was applied to the Logistic Regression model to identify which radiomics features most strongly influenced malignancy predictions. Summary plots visualize the distribution of SHAP values across the test set, revealing both feature importance and directional effects.

2.6 Bias Audit Framework

Aggregate performance metrics can mask significant disparities in model performance across subgroups—a critical concern in healthcare AI where such disparities could lead to differential care quality (Obermeyer et al., 2019). A systematic bias audit was conducted across three dimensions:

1. **Image Brightness:** Images were binned by mean pixel intensity (Dark: 0-80, Med-Dark: 81-95, Med-Light: 96-110, Light: 111-255) to assess whether image quality affects model accuracy
2. **Lesion Size:** Lesions were categorized by pixel area (Small: <1000, Med-Small: 1000-5000, Med-Large: 5000-15000, Large: >15000) to evaluate early detection capability
3. **Per-Class Performance:** Precision, recall, and F1-score were computed separately for each class with particular focus on malignant recall as the safety-critical metric

3. Results

3.1 Model Performance Comparison

Fifteen models were trained and evaluated across CNN and tabular paradigms. Table 1 presents the top performers, with complete results available in Appendix A.

Table 1: Top-Performing Models by F1-Score

Model	Type	Accuracy	F1-Score	AUC-ROC	Time (s)
Logistic Regression	Tabular (2-class)	93.75%	93.54%	0.952	0.01
XGBoost	Tabular (2-class)	93.75%	93.54%	0.984	0.64
ResNet-50	CNN (2-class)	89.69%	89.84%	0.959	255.00
ResNet-50 (Deployed)	CNN (3-class)	84.62%	84.86%	0.953	304.00
EfficientNet-B0	CNN (3-class)	82.91%	83.42%	0.937	285.00

Key Finding: Traditional machine learning models using hand-crafted radiomics features outperformed deep learning CNNs by approximately 4 percentage points in F1-score on 2-class classification. Logistic Regression achieved the highest F1-score (93.54%), demonstrating that expert-designed features remain highly effective for medical imaging tasks. However, for the educational application requiring 3-class classification and visual explanations, ResNet-50 was selected as the deployment model due to its Grad-CAM compatibility.

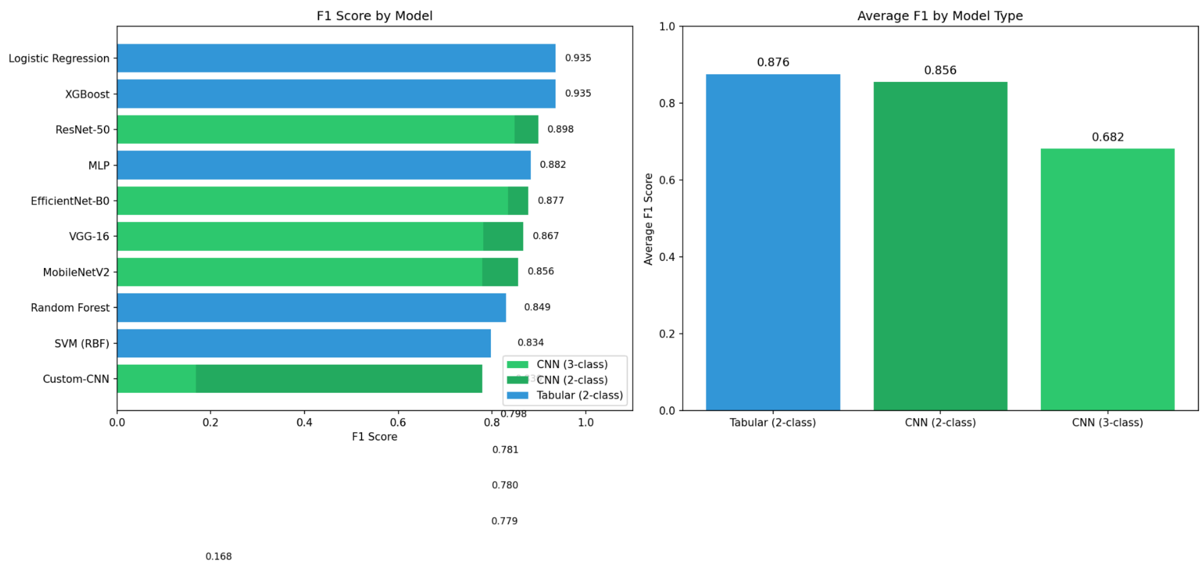

Figure 2: Model Performance Comparison (F1-Score)

3.2 Deployed Model Performance (ResNet-50 3-Class)

The ResNet-50 model trained for 3-class classification achieved 84.62% overall accuracy on the test set. Table 2 presents the per-class performance metrics.

Table 2: ResNet-50 Per-Class Performance on Test Set

Class	Precision	Recall	F1-Score	Support
Normal	79.2%	95.0%	86.4%	20
Benign	94.7%	81.8%	87.8%	66
Malignant	72.2%	83.9%	77.6%	31

Interpretation:

- **Normal (95.0% recall):** The model rarely misses healthy tissue, minimizing unnecessary concern for patients with normal scans
- **Benign (94.7% precision):** When the model predicts benign, it is highly reliable, reducing false reassurance for malignant cases
- **Malignant (83.9% recall):** The model catches 83.9% of malignant cases, though this falls short of the 90% target established in the success metrics
-

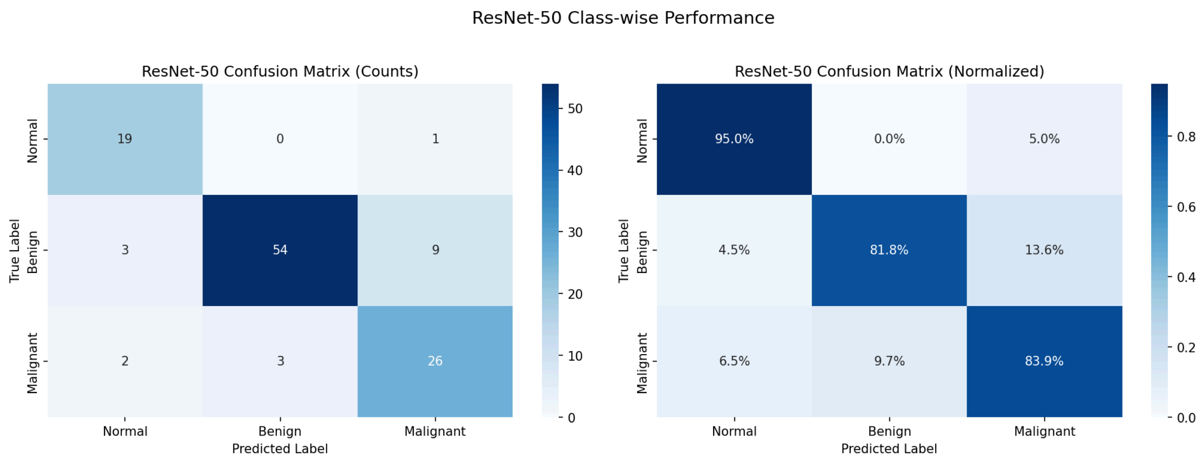

Figure 3: Confusion Matrix for ResNet-50 (3-Class) on Test Set

3.3 Explainability Analysis

3.3.1 Grad-CAM Visual Explanations

Grad-CAM analysis was performed on ResNet-50 to visualize which image regions influenced predictions. Figure 4 presents representative examples across all three classes.

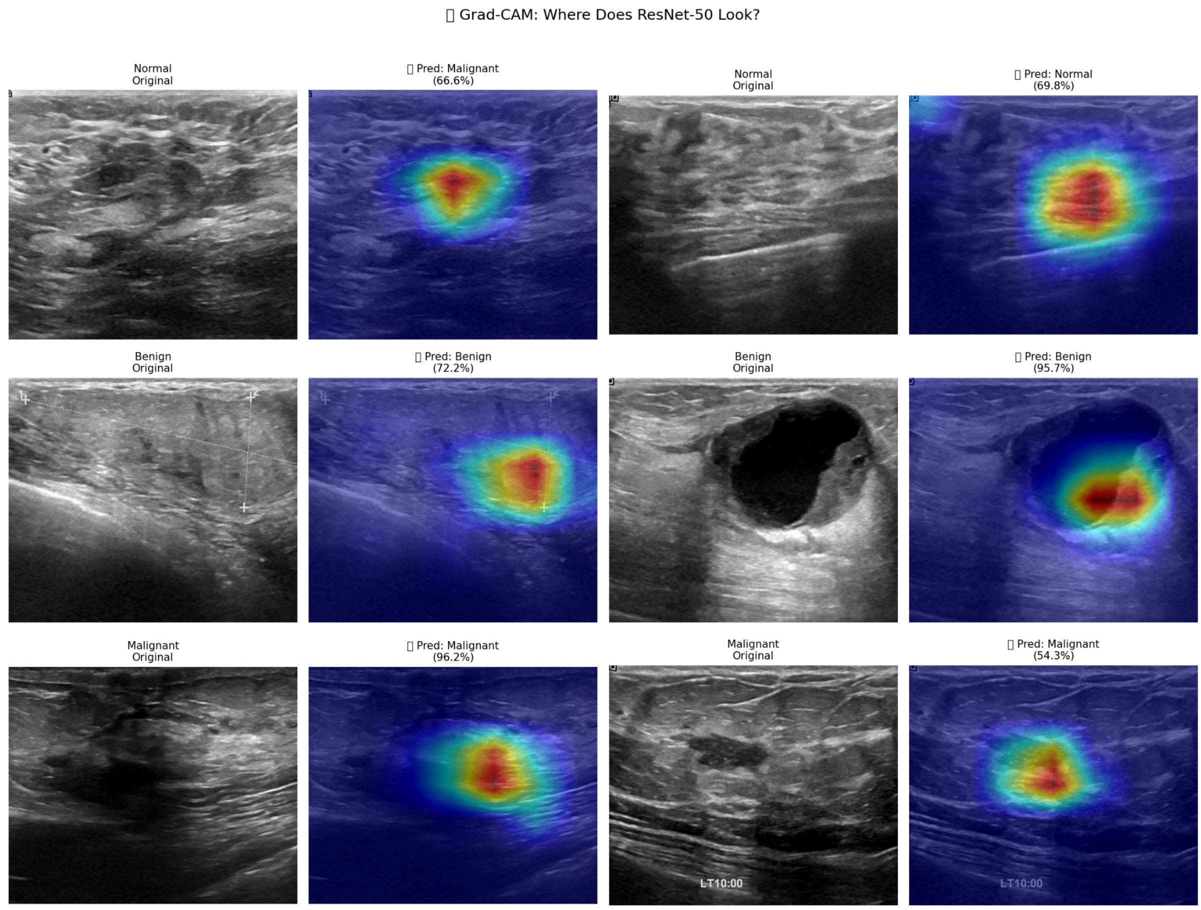

Figure 4: Grad-CAM Heatmaps Showing Model Attention for Each Class

Quantitative Grad-CAM Metrics:

- **Average Confidence:** 88.4% on correctly classified test samples
- **Average Focus Score:** 4.66, indicating concentrated attention on lesion regions rather than diffuse background activation
- **Clinical Alignment:** Heatmaps consistently highlight lesion boundaries and internal heterogeneity, consistent with features radiologists examine when applying BI-RADS criteria (D'Orsi et al., 2013)

The Grad-CAM visualizations provide evidence that the model attends to clinically meaningful regions rather than learning spurious correlations from image artifacts or background patterns. This alignment with radiological practice supports the model's use in educational contexts where understanding the 'why' behind predictions is essential for learning.

3.3.2 SHAP Feature Importance Analysis

SHAP analysis was applied to the Logistic Regression model to identify which radiomics features most strongly influence malignancy predictions. Table 3 presents the top 10 features ranked by mean absolute SHAP value.

Table 3: Top 10 Radiomics Features by SHAP Importance

Rank	Feature	Category	Effect Direction
1	shape2D_Sphericity	Shape	Low sphericity → Malignant
2	shape2D_Elongation	Shape	High elongation → Malignant
3	shape2D_Perimeter	Shape	Large/irregular → Malignant
4	shape2D_MinorAxisLength	Shape	Asymmetry indicator
5	glrlm_RunEntropy	Texture	High entropy → Malignant
6	shape2D_MajorAxisLength	Shape	Size indicator
7	firstorder_Entropy	First-Order	High complexity → Malignant
8	glcm_Contrast	Texture	High contrast → Malignant
9	shape2D_MaximumDiameter	Shape	Size indicator
10	glszm_ZoneEntropy	Texture	Heterogeneity measure

Clinical Validation of SHAP Features: The identified features align remarkably well with established BI-RADS lexicon descriptors for breast ultrasound (D'Orsi et al., 2013). The BI-RADS criteria identify irregular shape (low sphericity), non-parallel orientation (high elongation), non-circumscribed margins (irregular perimeter), and heterogeneous internal echotexture (high entropy) as features suggesting malignancy. This concordance between data-driven SHAP importance and expert-defined criteria provides strong evidence that the model has learned clinically meaningful patterns.

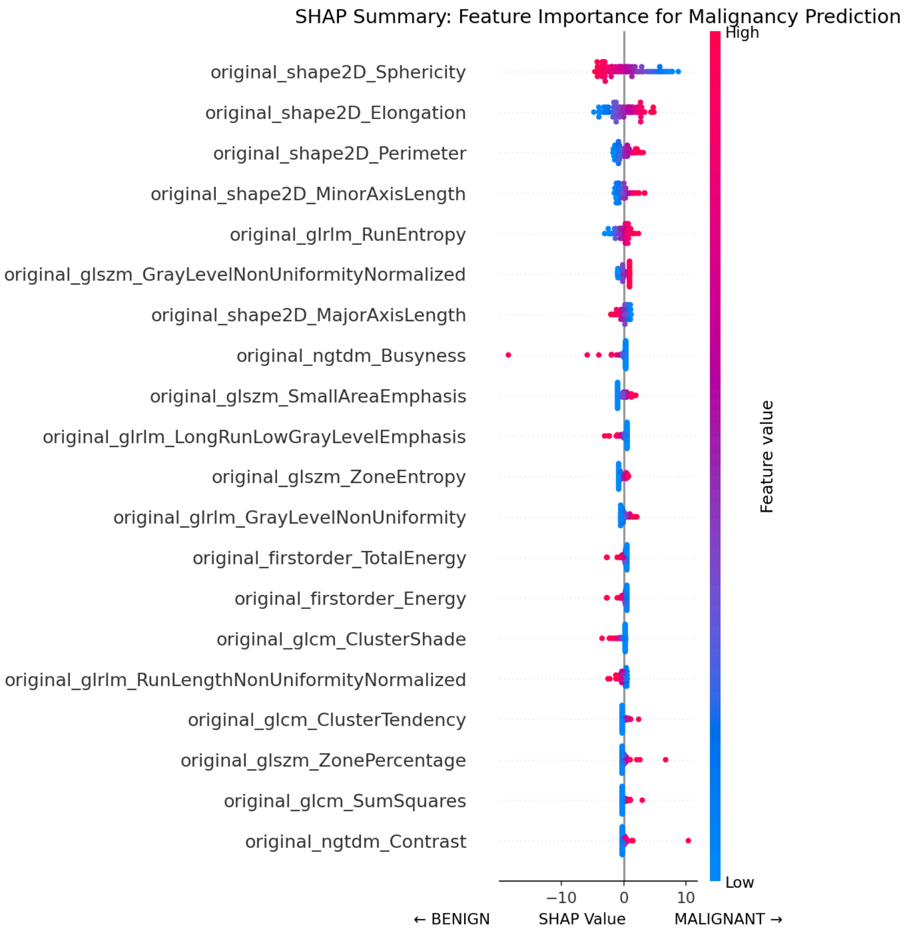

Figure 5: SHAP Summary Plot for Logistic Regression Model

3.4 Bias Audit Results

Systematic bias auditing revealed important performance variations across subgroups that would not be apparent from aggregate metrics alone.

3.4.1 Critical Finding: Small Malignant Lesion Detection

The most significant finding concerns small malignant lesion detection. While the model achieves 87.5% overall accuracy on small lesions, stratifying by class reveals a critical disparity:

Small Lesion Class	Recall	Correctly Identified	Status
Benign	95.0%	19/20	Excellent
Malignant	**50.0%**	2/4	**CRITICAL**

Clinical Implications: Small lesions often represent early-stage cancers where detection provides the greatest survival benefit. A 50% recall rate means half of small malignant lesions would be missed by the model—an unacceptable rate for any clinical application. However, for an educational tool with clear disclaimers, this limitation underscores the importance of professional oversight and serves as a teaching point about AI limitations.

3.4.2 Size and Brightness Bias Summary

Table 4: Bias Audit Results by Subgroup

Subgroup	Accuracy	Malignant Recall	Notes
Small Lesions	87.5%	50.0%	Critical gap for early detection
Med-Small Lesions	83.3%	N/A*	Moderate performance
Med-Large Lesions	87.5%	N/A*	Good performance
Large Lesions	70.8%	N/A*	Poor accuracy, fewer samples
Dark Images	86.7%	N/A*	Handles low quality well
Med-Dark Images	86.2%	N/A*	Good performance
Med-Light Images	75.9%	N/A*	Unexpected dip
Light Images	89.7%	N/A*	Best performance
Overall	84.6%	83.9%	Below 90% target

***N/A Explanation:** Malignant recall is marked as N/A for individual subgroups (Large Lesions, Med-Light Brightness, etc.) because the test set contained insufficient malignant samples within these specific bins to calculate a statistically meaningful recall rate. The bias audit stratifies first by image characteristic (size or brightness), then by class. When a particular bin contains zero or very few malignant cases, the recall calculation becomes undefined or unreliable. This limitation reflects the small dataset size (117 test images total, 31 malignant) rather than a methodological gap.

3.4.3 Evaluation Against Success Metrics

The project established a malignant recall target of >90% based on clinical CAD system standards. The achieved malignant recall of 83.9% falls approximately 6 percentage points below this target.

Success Metric Evaluation:

Metric	Target	Achieved	Status
F1-Score	> 80%	84.86%	✓ MET
Malignant Recall	> 90%	83.9%	✗ NOT MET (-6.1%)
AUC-ROC	> 0.90	0.953	✓ MET

Implications of Missing the Malignant Recall Target: The 83.9% malignant recall, while reasonable for an educational prototype, would be insufficient for a clinical screening tool. Approximately 16% of malignant cases (5 out of 31 in the test set) were misclassified—3 as benign and 2 as normal. In a clinical context, these false negatives could lead to delayed diagnosis. This shortfall reinforces the project's positioning as an educational tool only and highlights the gap between research prototypes and deployment-ready clinical AI systems. The limitation is prominently disclosed within the application interface.

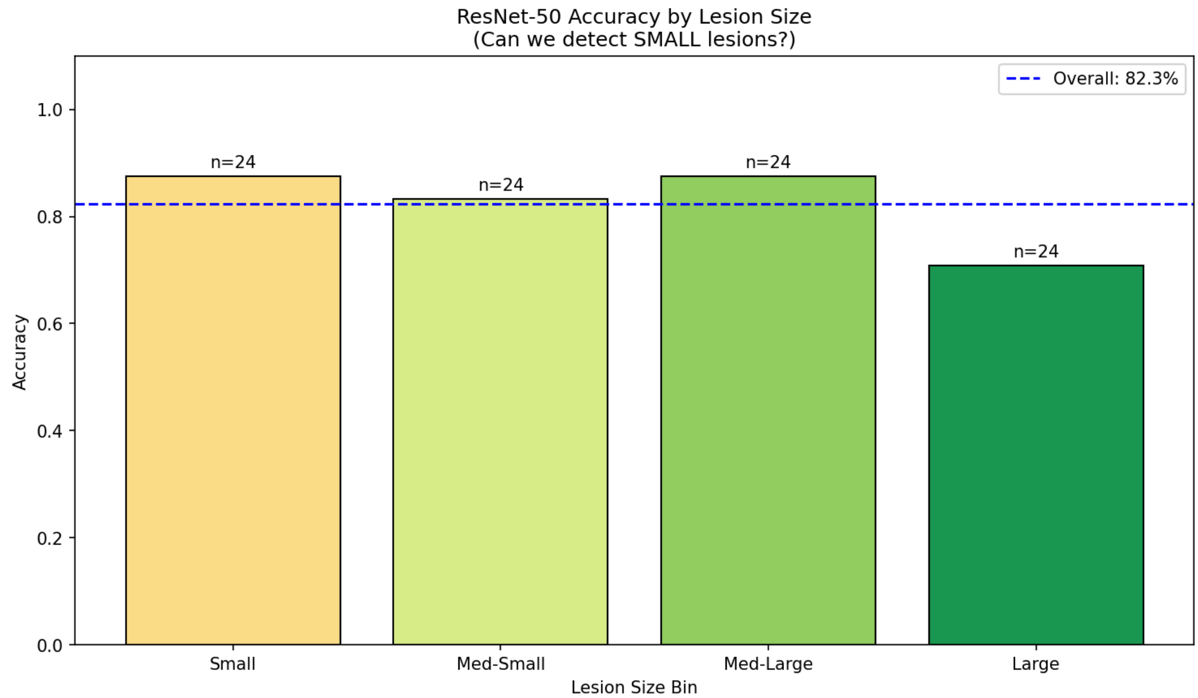

Figure 6: Model Accuracy Stratified by Lesion Size

4. Discussion

4.1 Key Findings

Radiomics vs. Deep Learning: One of the most notable findings is that traditional machine learning models using hand-crafted radiomics features outperformed deep learning CNNs by approximately 4 percentage points in F1-score. Logistic Regression achieved 93.54% F1-score compared to ResNet-50's 89.84% on binary classification. This finding aligns with recent literature suggesting that for small medical imaging datasets, expert-designed features can outperform learned representations (Afshar et al., 2019). The radiomics approach leverages decades of clinical knowledge encoded in features like sphericity and texture entropy, while deep learning must rediscover these patterns from limited data.

Explainability Validation: The concordance between SHAP-identified important features and BI-RADS criteria provides compelling evidence that the models learn clinically meaningful patterns. Shape features (sphericity, elongation, perimeter) dominate the importance rankings, consistent with the emphasis on lesion morphology in radiological assessment. Similarly, Grad-CAM heatmaps consistently highlight lesion boundaries and internal heterogeneity rather than background artifacts. This alignment is crucial for educational deployment, as users can trust that model explanations reflect genuine clinical reasoning.

Bias Audit Value: The systematic bias audit revealed critical performance disparities that would be invisible in aggregate metrics. The 50% recall on small malignant lesions is particularly concerning given that small lesions often represent early-stage disease where detection provides the greatest benefit. Without subgroup analysis, this dangerous gap would have gone undetected. This finding underscores the importance of comprehensive fairness evaluation in medical AI, as mandated by emerging regulations (FDA, 2021).

4.2 Limitations

- **Dataset Size and Diversity:** The BUSI dataset contains 780 images from a single institution in Egypt. This limited size and geographic scope may affect generalizability to other populations, imaging equipment, and acquisition protocols.
- **Small Malignant Lesion Performance:** The 50% recall on small malignant lesions is clinically unacceptable for any diagnostic application. This limitation must be prominently disclosed.
- **Malignant Recall Below Target:** The 83.9% malignant recall falls 6.1 percentage points below the 90% target, meaning approximately 1 in 6 cancers is missed.
- **No External Validation:** The model has not been validated on external datasets or against radiologist interpretations.
- **Subgroup Sample Sizes:** Some bias audit subgroups contained few samples, limiting the statistical power of disparity estimates.

4.3 Ethical Considerations and Safeguards

AMEEMAW is explicitly designed as an educational tool, not a diagnostic system. Several safeguards ensure responsible deployment:

- Prominent disclaimers on all screens stating the tool is not for clinical diagnosis
- Transparent disclosure of model accuracy (84.6%) and malignant recall (83.9%)
- Specific warning about 50% small malignant lesion detection rate
- Nana companion designed to educate and comfort, never to diagnose
- Consistent encouragement to consult healthcare professionals for all results

4.4 Future Work

- Expand dataset with multi-center data to improve generalizability across populations and equipment
- Implement attention mechanisms specifically targeting small lesion detection
- Conduct clinical validation study comparing model outputs to radiologist assessments
- Develop uncertainty quantification to flag low-confidence predictions for review
- Explore multi-modal approaches combining ultrasound with clinical history

5. Conclusion

AMEEMAW demonstrates the feasibility of developing educational AI tools for breast ultrasound interpretation while maintaining rigorous standards for transparency and responsible deployment. The comprehensive evaluation of 15 models revealed that radiomics-based Logistic Regression achieved the highest performance (93.54% F1-score) for binary classification, while ResNet-50 provided the best CNN performance (84.86% F1-score) with visual explainability through Grad-CAM.

Explainability analysis confirmed that both the CNN (via Grad-CAM) and tabular models (via SHAP) learn clinically meaningful patterns aligned with established BI-RADS criteria. The top radiomics features—sphericity, elongation, and perimeter—directly correspond to morphological descriptors used by radiologists to assess malignancy risk.

Critical bias auditing revealed significant performance limitations, most notably only 50% recall on small malignant lesions. This finding, along with the 83.9% overall malignant recall (below the 90% target), underscores the importance of subgroup analysis in medical AI and reinforces the tool's positioning as educational only. These limitations are prominently disclosed within the application.

The integration of Nana, a generative AI companion powered by Claude Haiku API, demonstrates how empathetic AI can support patients navigating the emotional aspects of breast health. By combining technical accuracy with compassionate communication, AMEEMAW aims to increase breast health awareness while maintaining appropriate boundaries between education and diagnosis.

This tool is intended for EDUCATIONAL PURPOSES ONLY and should never replace professional medical evaluation.

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Appendix A: Complete Model Comparison

#	Model	Type	Acc	Prec	Rec	F1	AUC
1	Logistic Regression	Tab (2c)	93.8%	94.3%	93.8%	93.5%	0.95
2	XGBoost	Tab (2c)	93.8%	94.3%	93.8%	93.5%	0.98
3	ResNet-50	CNN (2c)	89.7%	90.3%	89.7%	89.8%	0.96
4	MLP	Tab (2c)	88.5%	88.5%	88.5%	88.2%	0.90
5	EfficientNet-B0	CNN (2c)	87.6%	87.9%	87.6%	87.7%	0.95
6	VGG-16	CNN (2c)	86.6%	86.7%	86.6%	86.7%	0.95
7	MobileNetV2	CNN (2c)	85.6%	85.6%	85.6%	85.6%	0.95
8	ResNet-50	CNN (3c)	84.6%	86.1%	84.6%	84.9%	0.95
9	EfficientNet-B0	CNN (3c)	82.9%	86.3%	82.9%	83.4%	0.94
10	Random Forest	Tab (2c)	83.3%	83.0%	83.3%	83.0%	0.89
11	SVM (RBF)	Tab (2c)	82.3%	85.9%	82.3%	79.8%	0.96
12	VGG-16	CNN (3c)	77.8%	80.4%	77.8%	78.1%	0.92
13	Custom-CNN	CNN (2c)	77.3%	80.2%	77.3%	78.0%	0.86
14	MobileNetV2	CNN (3c)	77.8%	81.1%	77.8%	77.9%	0.94
15	Custom-CNN	CNN (3c)	27.4%	12.2%	27.4%	16.8%	0.67

Note: Tab = Tabular, CNN = Convolutional Neural Network, 2c = 2-class (Benign/Malignant), 3c = 3-class (Normal/Benign/Malignant)